Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 04 · Part B: Mathematics of Signals and Systems Level 2

Track: Fourier Series, Fourier Transform, Gibbs Phenomenon, Uncertainty Principle, Harmonic Analysis

Prerequisites: Problems 02, 03 · **Estimated effort:** 10–14 hours

Fourier Series and the Fourier Transform: Seeing a Signal as a Sum of Frequencies

Derive the Fourier coefficients of the standard waveforms by hand, watch partial sums converge (and overshoot), prove numerically that no signal can be short in both time and frequency, and read a rotating machine's vibration spectrum to extract shaft speed and blade-pass frequency.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/04_fourier_series_and_transform/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

The Fourier transform is the lens through which almost all instrument data is examined. A vibration record that looks like noise in time shows sharp lines at multiples of the shaft speed; a clipped amplifier output reveals itself as odd harmonics; a pulse of finite length has a spectrum whose width is set by its duration. Convolution in time becomes multiplication in frequency, which turns filtering and system analysis into algebra.

Here you work in the continuous-time picture (approximated on a fine grid) so that the theory and numerics can be compared term by term. The discrete transform (DFT/FFT), with all its practical pitfalls, is the subject of problem 12; this problem prepares you by showing, with controlled examples, what the transform *should* give.

Theoretical Framework

1. Fourier Series

A periodic signal of period T_0 ($\omega_0 = 2\pi/T_0$) has

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \, \mathrm{e}^{\mathrm{j}k\omega_0 t}, \qquad c_k = \frac{1}{T_0}\int_0^{T_0} x(t)\, \mathrm{e}^{-\mathrm{j}k\omega_0 t}\, \mathrm{d}t, \qquad P_x = \frac{1}{T_0}\int_0^{T_0} |x|^2 \mathrm{d}t = \sum_k |c_k|^2 \ \text{(Parseval)}. \tag{1}$$

For a real signal $c_{-k} = c_k^*$ and the trigonometric form is $x = a_0 + \sum_{k\geq 1}(a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$ with $|c_k| = \frac{1}{2}\sqrt{a_k^2 + b_k^2}$. The unit-amplitude odd square wave has $b_k = 4/(\pi k)$ for odd k (and zero otherwise); the unit triangle wave has $b_k = \pm 8/(\pi^2 k^2)$ for odd k; the sawtooth has $b_k = \mp 2/(\pi k)$. A discontinuity gives coefficients decaying as $1/k$, a continuous signal with a corner as $1/k^2$.

2. Gibbs Phenomenon

Partial sums of a function with a jump overshoot the jump by about 8.95% *of the jump height*, no matter how many terms are kept; the overshoot narrows but does not shrink. Smoothing the coefficients with Fejér (Cesàro) weights $(1 - |k|/(K+1))$ or Lanczos σ-factors removes it at the cost of a slower edge.

3. Fourier Transform and Its Properties

$$X(f) = \int_{-\infty}^{\infty} x(t)\, \mathrm{e}^{-\mathrm{j}2\pi f t}\mathrm{d}t, \qquad x(t) = \int_{-\infty}^{\infty} X(f)\, \mathrm{e}^{\mathrm{j}2\pi f t}\mathrm{d}f, \qquad \int |x|^2\mathrm{d}t = \int |X|^2\mathrm{d}f. \tag{2}$$

Key pairs: $\mathrm{rect}(t/T) \leftrightarrow T\,\mathrm{sinc}(fT)$; $\mathrm{e}^{-t^2/2\sigma^2} \leftrightarrow \sigma\sqrt{2\pi}\,\mathrm{e}^{-2\pi^2\sigma^2 f^2}$; $\mathrm{e}^{-a|t|} \leftrightarrow 2a/(a^2{+}4\pi^2 f^2)$; an impulse train of period $T \leftrightarrow$ an impulse train of spacing $1/T$. Properties: shift $x(t-t_0) \leftrightarrow X(f)\mathrm{e}^{-\mathrm{j}2\pi f t_0}$; modulation $x(t)\cos 2\pi f_c t \leftrightarrow \frac{1}{2}[X(f-f_c)+X(f+f_c)]$; scaling $x(at) \leftrightarrow X(f/a)/|a|$; and $x * h \leftrightarrow XH$.

4. Time–Bandwidth Uncertainty

Define the RMS durations $\Delta t^2 = \int t^2|x|^2\mathrm{d}t / \int |x|^2\mathrm{d}t$ and $\Delta f^2 = \int f^2|X|^2\mathrm{d}f / \int |X|^2\mathrm{d}f$ (signal centred at zero). Then

$$\Delta t\, \Delta f \;\geq\; \frac{1}{4\pi}, \tag{3}$$

with equality only for the Gaussian pulse. A signal cannot be both short and narrow-band; this is the origin of the window trade-offs of problems 12 and 17.

5. Total Harmonic Distortion

For a signal with fundamental amplitude A_1 and harmonics $A_2, A_3, \ldots$, $\mathrm{THD} = \sqrt{\sum_{k\geq 2} A_k^2}/A_1$, often quoted in dB or percent.

Data and Simulation Specification

Waveforms	Unit-amplitude square, triangle and sawtooth waves, $T_0 = 1\,\mathrm{s}$; sine clipped at ± 0.7 of its amplitude for the THD task.
Numerics	Dense grid $\Delta t \leq 0.1\,\mathrm{ms}$; evaluate c_k and $X(f)$ by trapezoidal integration (not by FFT) first, then compare with FFT scaling $X(f_k) \approx \Delta t\,\mathrm{FFT}[k]$.
Pulses	Rectangular ($T = 20\,\mathrm{ms}$), Gaussian ($\sigma = 5\,\mathrm{ms}$), two-sided exponential ($a = 100/\mathrm{s}$), linear chirp ($100\,\mathrm{Hz} \to 400\,\mathrm{Hz}$ in $200\,\mathrm{ms}$).
Machine signal	`sp_common.synth.bearing_vibration(fault="none", shaft_hz=29.95, fs=12000, duration=10)`: shaft harmonics plus noise.
Real data (choose one)	(i) Record a tuning-fork app, your voice saying "aaah" or a running fan with Phyphox (audio or accelerometer). (ii) Monthly mean sunspot number from SILSO (Royal Observatory of Belgium): find the solar-cycle period by Fourier analysis. (iii) Normal-baseline drive-end record of the Case Western Reserve University Bearing Data Center (12 kHz sampling).

Your Task

1. **Fourier series by hand and by code (Stage 2).** Derive c_k for the square, triangle and sawtooth waves on paper. Compute them by numerical integration, plot $|c_k|$ against k on log–log axes and fit the decay exponents (-1, -2, -1). Plot partial sums with $K = 1, 3, 5, 11, 51$ harmonics and the RMS error versus K.
2. **Gibbs phenomenon.** Measure the overshoot of the square-wave partial sums for K up to 1001 and show it converges to about 17.9% of the half-amplitude (8.95% of the jump). Apply Fejér and Lanczos smoothing and compare the overshoot and the 10–90% edge width.
3. **Parseval and distortion.** Verify Eq. (1) for the square wave (the sum of $|c_k|^2$ must tend to 1). Compute the THD of the clipped sine analytically from its Fourier coefficients and numerically;

plot THD versus clipping level.

4. **Transform pairs.** Compute $X(f)$ numerically for the rectangle, Gaussian and exponential pulses, overlay the analytic transforms, and report the maximum relative error. Show what happens when the integration window truncates the exponential or the sampling step is too coarse.
5. **Properties.** Verify time shift (phase ramp), modulation (spectrum shift to $\pm f_c$, i.e. AM), scaling (time compression widens the spectrum) and the convolution theorem (filtering by multiplying in frequency versus convolving in time) numerically.
6. **Uncertainty principle.** Measure $\Delta t\,\Delta f$ for the Gaussian, rectangle, exponential and chirp pulses and confirm Eq. (3), with equality only for the Gaussian. Show how the Gaussian's product changes if you apply a chirp phase.
7. **Rotating machinery.** Compute the amplitude spectrum of the machine signal and of your real recording. Identify the harmonic series, estimate the shaft frequency from the harmonic spacing and convert to rpm.

Self-check targets

Square wave: $b_1 = 1.2732$, $b_3 = 0.4244$, and $\sum |c_k|^2 \to 1$. Triangle-wave coefficients fall as $1/k^2$. The Gibbs peak of a unit square wave's partial sum approaches 1.179. For the Gaussian, $\Delta t\,\Delta f = 1/(4\pi) = 0.0796$; for the rectangle it is larger (and for the rectangle Δf is not finite, so use a truncated frequency band and say so). The shaft of the synthetic machine is 29.95 Hz, i.e. 1797 rpm.

Aerospace Application Link

A two-blade propeller turning at 4800 rpm has a shaft frequency of 80 Hz and a blade-pass frequency of 160 Hz; airframe panels excited at those lines (and their harmonics) fatigue. Using the machine spectrum, state which lines you would monitor on a flight test for an engine with N_b blades, compute the THD of the strain-gauge amplifier that clips at 70% of full scale, and decide whether the amplifier gain must be reduced to keep THD below 1%.

Deliverables

D1 **Fourier series notebook:** hand derivations (scanned or typeset), coefficient plots with fitted exponents, partial sums and error versus K.

D2 **Gibbs study:** overshoot versus K, Fejér and Lanczos comparison.

D3 **Parseval and THD:** the power sum and the THD-versus-clipping plot.

D4 **Transforms and properties:** numerical versus analytic transforms with errors, and the four property demonstrations.

D5 **Uncertainty and machine spectrum:** table of $\Delta t\,\Delta f$ values and the annotated machine spectrum with rpm estimate.

D6 **Memo (250 words):** "What can the spectrum tell us about the health of a rotating machine?" Cite at least two external sources.

Bonus (a) Show that the Fourier transform of a Gaussian pulse is again Gaussian by evaluating the integral analytically (complete the square). (b) Compute the transform of a pulse train and explain the line spacing $1/T$. (c) Compute the Hilbert transform of a tone and the analytic signal of an AM waveform.

Learning Resources

Fourier Analysis

- Oppenheim, A.V., Willsky, A.S. *Signals and Systems*, Ch. 3 (Fourier series) and Ch. 4 (continuous-time Fourier transform).
- Bracewell, R.N. *The Fourier Transform and Its Applications* (pairs, theorems, uncertainty relation).
- Smith, S.W. *The Scientist and Engineer's Guide to Digital Signal Processing*, Ch. 8–10; https://www.dspguide.com.

Machinery and Data

- Randall, R.B. *Vibration-Based Condition Monitoring* (spectra of rotating machines).
- Smith, W.A., Randall, R.B. (2015). "Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study." Mech. Syst. Signal Process. 64–65, 100.
- CWRU Bearing Data Center: https://engineering.case.edu/bearingdatacenter; SILSO sunspot data: https://www.sidc.be/SILSO/

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.